Using Apache Iceberg for Developing Modern Data Tables

Apache Iceberg, a data management powerhouse, empowers businesses with its remarkable flexibility, seamless tool compatibility, and endorsement by the major tech players in the industry.

The cloud has made it possible to collect vast quantities of data and store it at a reasonable cost. It is also responsible today for the emergence of data lakes, data mesh and other modern data architectures. However, handling large amounts of data using a generic cloud offers its own challenges and limitations.

For instance, query engines often cannot react with typical blob storage systems smoothly. To address this issue, applying a table format to data becomes extremely useful.

Table format data was originally developed by Netflix and was open sourced as an Apache incubator project in 2018. It graduated from this in 2020 and became Apache Iceberg. Apache Iceberg can meet certain challenges faced by Apache Hive.

Process engines used by a company may change from time to time. For example, many business firms have moved from Hadoop to Spark or Trino. Iceberg provides greater flexibility and choice for processing engines.

Additionally, data lakes have undesirable qualities due to table formats obtained from older technologies. Apache Iceberg is built to address such shortcomings in Apache Hive. Moreover, Iceberg also supports multiple file formats, including Apache Parquet, Apache Avro and Apache ORC. Hence it provides more flexibility and long-term adaptability for file formats that may emerge in the future.

Reasons to choose Apache Iceberg

Apache Iceberg is a new open table format developed by the open source community. The modern data industry needs many metadata tables such as data files tables, all manifest tables, all entries tables, history tables, partition tables, and snapshot tables, etc. Iceberg provides an effective solution for the creation of such tables.

Table 1 compares several features of three major data lake table format creators: Apache Iceberg, Apache Hudi and Delta Lake.